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Predicting the critical temperature of superconductors and materials classification with a balanced dataset without prior knowledge

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P. A. H. Nawoda,^{1,a)}  Gayan P. Hettiarachchi,^{2,3,b),c)}  and M. D. T. Attygalle^{1,d)} 

AFFILIATIONS

¹Department of Statistics, University of Colombo, College House, 94 Kumaratunga Munidasa Mawatha, Colombo 00700, Sri Lanka

²R3 Institute for Newly-Emerging Science Design, Osaka University, 1-2 Machikaneyama, Toyonaka, Osaka 560-0043, Japan

³Department of Digital Business and Innovation, Tokyo International University, 4-42-31 Higashi-Ikebukuro, Toshima, Tokyo, 170-0013 Japan

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^{a)}E-mail: 2019s17415@stu.cmb.ac.lk

^{b)}Author to whom correspondence should be addressed: ghettiar@tiu.ac.jp

^{c)}Present/Permanent Address: Department of Digital Business and Innovation, Tokyo International University, 4-42-31 Higashi-Ikebukuro, Toshima, Tokyo 170-0013, Japan.

^{d)}E-mail: dilhari@stat.cmb.ac.lk

ABSTRACT

The discovery of new superconductors through traditional trial-and-error experimental methods is both challenging and costly. A data-driven strategy can speed up the time-consuming exploration processes and also potentially reveal new insights into the complex correlations at play. Several material databases have been created for predicting the superconducting critical temperature (T_c), but they are confined solely to superconducting materials, limiting their prediction capabilities. In addressing this, we compiled a comprehensive open-source dataset encompassing both superconductors and non-superconductors. Our dataset contains 13 415 superconductors and 13 425 non-superconductors, each characterized by 212 features. Novel features, namely, CuO layers, ionic radius, heat of vaporization, cohesive energy, and thermal conductivity with respect to every element in the material were included. Another feature named “material type” was introduced to classify materials as low-temperature superconductors, high-temperature superconductors, and non-superconductors. Various machine learning techniques, including boosting and bagging methods, were employed to predict T_c . The light gradient boosting model emerged as the most effective, achieving a coefficient of determination of 93% and a mean absolute error of 2.93 K. While this novel, comprehensive material dataset is made available to enrich future research, a web application is also developed for predicting T_c of any material in relation to the material type based on the best of trained models.

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I. INTRODUCTION

Ever since the discovery of superconductivity by Onnes in 1911,¹ zealous research efforts have revealed many superconducting elements and materials below their so-called superconducting critical temperature (T_c).² These elements and materials demonstrate a wide

range of T_c values and are often divided into high-temperature superconductors (HTS) having a T_c higher than 77 K (above the boiling point of liquid nitrogen) and low-temperature superconductors (LTS) with a T_c lower than 77 K.³ Such superconducting materials, especially HTS, play a critical role in transportation, medical, and electrical-power industries and have the potential to revolutionize

next-generation technology.^{4,5} Hence, the speedy and efficient discovery and design of HTS, preferably at ambient conditions, are extremely important from an engineering point of view. Furthermore, the discovery of novel superconductors can provide additional insight into understanding the detailed physics of the superconducting phenomenon and, hence, plays a critical role in the advancement of the basic sciences.

The discovery of HTS has traditionally relied on extensive experimental and computational techniques based on the first-principles.^{6,7} The entailing trial-and-error process often requires conducting experiments at extremely low temperatures and/or under very high pressures, making the search for new superconductors both challenging and resource-intensive.^{8,9} Moreover, the first-principles calculations based on the density functional theory (DFT), which are commonly used in computational research, tend to be time-consuming and costly.^{10,11} These limitations have slowed the design, discovery, and commercialization of HTS. Recently, advances in data-driven methodologies and machine learning (ML) techniques have opened up new avenues for overcoming these challenges. By leveraging large datasets and advanced predictive algorithms, researchers can now accelerate the identification, design, and optimization of HTS, making the search process more efficient and cost-effective. These innovative approaches hold promise for significantly advancing the field and facilitating the development of practical superconducting technologies.

In one such data-driven study, Owolabi *et al.*¹² used support vector machines (SVMs) to estimate T_c of a family of iron-based superconductors at ambient pressure and achieved above 99% accuracy. Stanev *et al.*¹³ conducted a study using a dataset of 16 413 materials, each characterized by 143 features from the Material-Agnostic Platform for Informatics and Exploration (Magpie), which is a flexible platform/tool designed to predict material properties using machine learning. The features were weighted elemental properties, such as electronegativity, melting point, and electron affinity. In their study, the materials were categorized based on T_c being above or below 10 K, and the classification model achieved an accuracy of 92% in predicting whether a material's T_c is above or below the 10 K threshold. Additionally, they developed separate regression models to predict T_c of cuprate, iron-based, and low- T_c superconducting compounds. These models offer valuable insights into the underlying mechanisms of superconductivity across different material families, which are extremely difficult to gauge experimentally and/or theoretically due to the cohort of microscopic parameters that play a critical role. The regression models yielded coefficient of determination (R^2)

values of approximately 0.85 for low- T_c materials, slightly below 0.80 for cuprates, and around 0.74 for iron-based compounds. Hamidieh¹⁴ utilized multiple regression with extreme gradient boosting (XGBoost) for regression tasks in order to perform statistical modeling in predicting T_c . The approach achieved an R^2 value of 0.92 and a root mean square error (RMSE) of 9.5, greatly outperforming multiple linear regression (MLR) models.

Agbemade¹⁵ used model diagnostics and the backward selection approach for feature selection, and after adjusting for hyperparameters the gradient boosted approach produced an R^2 value of approximately 0.88, indicating that it performed better than MLR.¹⁵ Liu *et al.*¹⁶ presented a technique that integrates principal component analysis (PCA), particle swarm optimization (PSO), and support vector regression (SVR). This PCA-PSO-SVR approach obtained an outstanding R^2 value of 0.98. In further expanding ML-based approaches, the straightforward chemical formula was used as the input by Gaikwad and Doke¹⁷ and a variety of regression techniques, such as Random Forest (RF), Decision Tree (DT), Bayes model, MLR, PCA, SVR, and XGBoost, were used.¹⁷ Shams *et al.*¹⁸ trained on the intricate properties of superconductive materials using a stacking ML technique and attained an RMSE of 9.68 and an R^2 value of approximately 0.92. ML techniques such as ridge regression, lasso regression, elastic net, and whale optimization algorithm/multivariate adaptive regression splines (WOA/MARS) were utilized by García-Nieto *et al.*¹⁹ Zhang *et al.*²⁰ used an integrated ML model to obtain a prediction accuracy of 95.9% through cross-validation employing Extra Trees (ET), Light Gradient Boosting Machine (LGBM), and Gradient Boosting Decision Trees (GBDT).

Table I provides a summary of the results obtained in past studies, including the size of the dataset and the achieved accuracy. This compilation offers insights into the performance of various models across different datasets, facilitating comparisons and informing future research directions.

Table I highlights that in the previous studies, different materials, features, and types of machine learning (ML) techniques were used in predicting T_c . As a result, these existing databases do not encompass all materials and common features due to the different data collection, data cleaning, and preprocessing techniques they have used. Moreover, the ML models are specifically trained only on superconductors. As is the case with much of the existing research, training a model exclusively on superconductors introduces bias, leading to inaccurate predictions when evaluating completely new materials. Such a model may fail to accurately

TABLE I. The summary of selected past studies highlighting the dataset, best performing model, and model performance.

Research	Dataset (materials, features)	Best model	R^2 value (%)
Owolabi <i>et al.</i> ¹²	(32, 2)	SVM	99.0
Hamidieh ¹⁴	(21 263, 81)	XGBoost	92.0
Agbemade ¹⁵	(21 263, 81)	Gradient Boosting (tuning)	88.2
Liu <i>et al.</i> ¹⁶	(31, 7)	PCA-PSO-SVR	98.0
Shams <i>et al.</i> ¹⁸	(21 263, 81)	Stacking	92.2
García-Nieto <i>et al.</i> ¹⁹	(21 263, 81)	WOA/MARS	80.0
Zhang <i>et al.</i> ²⁰	(13 138, 115)	ET + LGM + GBDT	95.9

distinguish whether a new material under scrutiny is a superconductor or a non-superconductor. Therefore, it is crucial to have a diverse and comprehensive dataset and to train models on varied materials to improve the accuracy of prediction tasks in data-driven approaches for materials design and superconductivity research.

In addressing this, we constructed a comprehensive dataset that includes both superconductors (13 415) and non-superconductors (13 425). In addition to the features considered in previous works, we introduced new features to the dataset that incorporate updated values of element-related properties and additional properties not generated by Magpie software nor considered in previous works.¹³ These additional properties include the ionic radius, cohesive energy, and heat of vaporization. Thermal conductivity is another critical factor in predicting T_c .³ While statistical measurements related to features are included, they often fail to capture individual element variations. To address this, our study incorporated element properties along with material stoichiometry for each element in a material, capturing the doping variations in the material. Our data also indicate whether T_c depends on the presence of CuO layers in the materials³ as a variable with appropriate weighting. The type of material, specifically, whether a material is LTS, HTS, or non-superconductor is also included. During the dataset compilation, materials with differing critical temperatures were handled by using the most recent value found in the literature, and we excluded permutations that differ only in how the chemical composition is denoted. Our analysis of this new dataset employed a comprehensive approach, considering all training and testing evaluation metrics to select the best model and ensure that overfitting was avoided. In comparison to previous studies that considered only superconductors in prediction tasks, our approach, which includes both superconductors and non-superconductors, can predict T_c of any given material with 93% accuracy, making it accessible to users with or without prior knowledge of the material's chemical properties.

II. METHODS

In predicting T_c in superconducting materials using ML, two fundamental steps are essential. First, it is necessary to gather and preprocess the dataset. Second, an appropriate algorithm must be selected for the learning process and model development using the dataset. Halevy *et al.* emphasize the importance of the initial step, noting that data scientists often spend around 80% of their efforts on data collection and preprocessing.^{21,22} This aspect is applicable in the current study as well, which utilizes the Supercon dataset,^{23,24} the largest and most comprehensive database of superconducting materials, that is further upgraded and enhanced with respect to both materials and encompassing features.

In this work, significant effort is dedicated to executing various data preprocessing steps, and detailed explanations are provided for each stage below. After implementing multiple phases of data preprocessing, and excluding problematic and incomplete data, a comprehensive dataset was created for developing an effective model for predicting T_c .

A. Data cleaning

Starting with the SuperCon dataset, consideration was initially placed on removing duplicate or irrelevant observations and

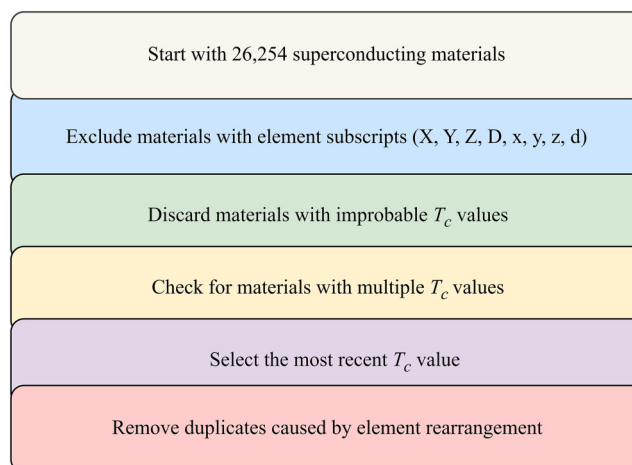


FIG. 1. The end-to-end procedure for cleaning the superconductor data obtained from the SuperCon dataset, which initially consisted of 26 254 superconducting materials.

unrealistic values of features, and fixing structural errors of the materials as described below. The initial SuperCon dataset consisted of 26 254 superconducting materials and 87 non-superconductors (only about 0.3% of data) before cleaning.

We excluded compounds with element subscripts such as X, Y, Z, D, x, y, z, and d. For instance, compounds like $\text{Sm}_1\text{Ba}_2\text{Cu}_{2.9}\text{Ni}_{0.1}\text{O}_{7-z}$, $\text{Y}_1\text{Ba}_2\text{Cu}_3\text{O}_{7-d}$, and $\text{Ru}_{0.6}\text{Sr}_2\text{Gd}_1\text{Cu}_{2.4}\text{O}_{8+z}$ were removed. Additionally, compounds containing elements not found in the periodic table are disregarded due to possible typographical errors. $\text{Y}_1\text{Ba}_2\text{Cu}_1\text{O}_{6050}$, for instance, is excluded because the oxygen subscript of 6050 appears to be incorrect. The compound $\text{Pb}_2\text{C}_1\text{Ag}_2\text{O}_6$ is discarded due to an improbable reported T_c of 323 K. Given the current state of research on room-temperature superconductors, this value is unrealistic for this material. A challenge within the SuperCon dataset is the presence of multiple reported T_c values for certain compounds, which hinders precise analysis. For instance, the compound Mg_1B_2 has been documented with varying transition temperatures over different years. To address this issue, we utilized the most recent T_c value available for each compound. Some compounds are considered duplicates due to the rearrangement of elements; for example, Mg_1B_2 and B_2Mg , $\text{Ag}_7\text{B}_1\text{F}_4\text{O}_8$ and $\text{Ag}_7\text{F}_4\text{O}_8\text{B}_1$, $\text{Al}_{0.1}\text{Si}_{0.9}\text{V}_3$ and $\text{V}_3\text{Si}_{0.9}\text{Al}_{0.1}$, and Zr_2Co_1 and Co_1Zr_2 . In such cases, we removed duplicates to avoid redundancy and improve efficiency in the learning process. Figure 1 illustrates the end-to-end procedure for cleaning the superconductor dataset.

B. Data integration

Non-superconductors are materials that do not exhibit superconductivity and remain metals or insulators down to very low temperature. Metals adhere to Ohm's law, which states that the current passing through them is directly proportional to the applied voltage.²⁵ The chemical composition and related properties of such

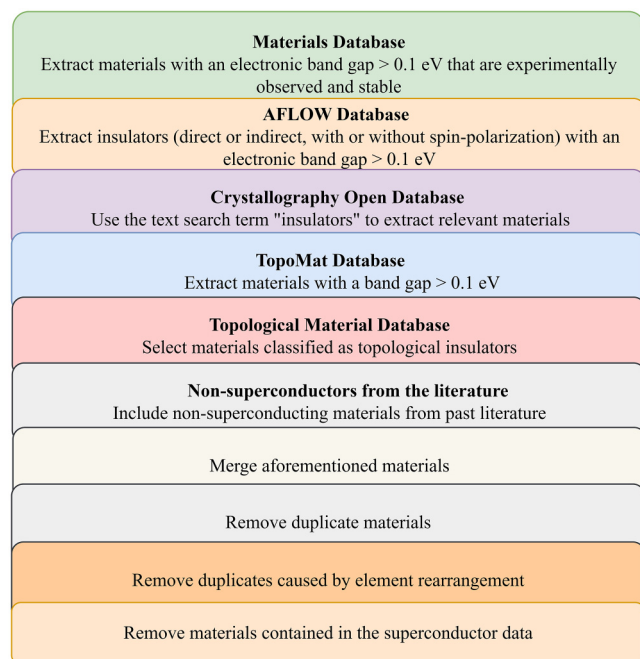


FIG. 2. Procedure for creating and cleaning the non-superconductor materials dataset used in this study. Combining this dataset with the superconductor materials dataset creates a balanced and comprehensive dataset for training the machine learning models.

non-superconducting materials exist in various databases. In this study, information on non-superconducting materials was integrated using several established databases, including the Crystallography Open Database (COD), AFLOW, Materials Project, Topomat Database, and Topological Materials Database. Additionally, datasets reviewed in the past literature were also utilized in the data-gathering and dataset creation processes.^{26–31} Figure 2 summarizes the procedure for creating and cleaning the non-superconductor materials dataset used in this study.

By integrating information from a variety of existing databases, this study creates comprehensive data on both superconductors and non-superconductors. Use of such a dataset ensures that the model can more accurately predict whether a new material is a superconductor or a non-superconductor, suppressing the bias introduced by training predictive models with only superconductors.

C. Material features

After constructing the dataset, the next step was to gather material-related features for the compounds. While attempting to add useful features using material databases, many were found to be unavailable, posing a challenge in obtaining valid experimental results. In addressing this, we utilized the Magpie software for feature derivation.^{32–35} However, this method did not encompass all the element-related features crucial for our prediction task. To address this limitation, we developed a dataset comprising the

periodic elements with their associated physical, chemical, electrical, and magnetic properties. Placeholder features were derived to represent actual features by considering statistical metrics such as the mean, standard deviation, mode, minimum, maximum, and range.

Several other element-related features considered in previous works, but not included in Magpie, are also useful for predicting T_c . Namely, we considered the atomic radius, electron affinity, valence, thermal conductivity, fusion heat, density, first ionization energy, and number of elements. In addition to these features, we considered several other element-related features such as the ionic radius, cohesive energy, and heat of vaporization that could potentially improve the predictive accuracy and are novel features.

In deriving the aforementioned features for a material, first the material's constituent elements and their corresponding stoichiometric coefficients were extracted. Subsequently, these were utilized in various statistical formulas to compute a set of features. For example, assuming that the element-related feature is the atomic mass, statistical measurement is taken as the mean based on the chemical formula and ratio. Other features, such as ionic radius, cohesive energy, heat of vaporization, electron affinity, and fusion heat, can be derived in a similar manner.

In further improving the dataset, important variables that play a critical role in determining T_c of materials were incorporated, for instance, the presence of layers that consist of copper (Cu) and oxygen (O). The presence of CuO layers can be quantified by the weight of Cu and O elements. Another novel feature included is the ability to capture doping variations and elemental variations in the material. This was achieved by calculating the thermal conductivity values for the stoichiometry of each element in the material, as well as considering the material type to identify different patterns. Material types include three categories, namely, LTS ($T_c < 77$ K), HTS ($T_c > 77$ K), and non-superconductors ($T_c = 0$ K). For instance, if a material has the chemical formula $\text{Ag}_{0.6}\text{Al}_{0.4}$ and thermal conductivities of silver (Ag) and aluminum (Al) are 450 and 235 W/m·K, respectively, the new features would consist of the thermal conductivity value of the first element (C1) as 0.6×450 and thermal conductivity value of the second element (C2) as 0.4×235 .

The improved dataset consists of 13 415 superconductors and 13 425 non-superconductors, ensuring a balanced representation of superconductors and non-superconductors. Each compound is characterized by 212 features, including three novel features facilitating a comprehensive analysis and improved prediction accuracy as discussed later.

D. Feature selection

Feature selection techniques are crucial for identifying the most relevant subset of features in a dataset for a prediction task. These techniques offer numerous benefits, including minimizing overfitting, enhancing accuracy, reducing training time, simplifying model design, achieving faster convergence, improving generalization, and increasing robustness to noise.^{36–38} In the context of superconducting materials, feature selection helps pinpoint the attributes most closely associated with T_c .

In this study, we employed an embedded feature selection approach³⁹ using a RF regression model to select the most relevant

TABLE II. The summary of ML techniques adopted in this study. The description highlights the advantage(s) of using the respective model in predictive tasks.

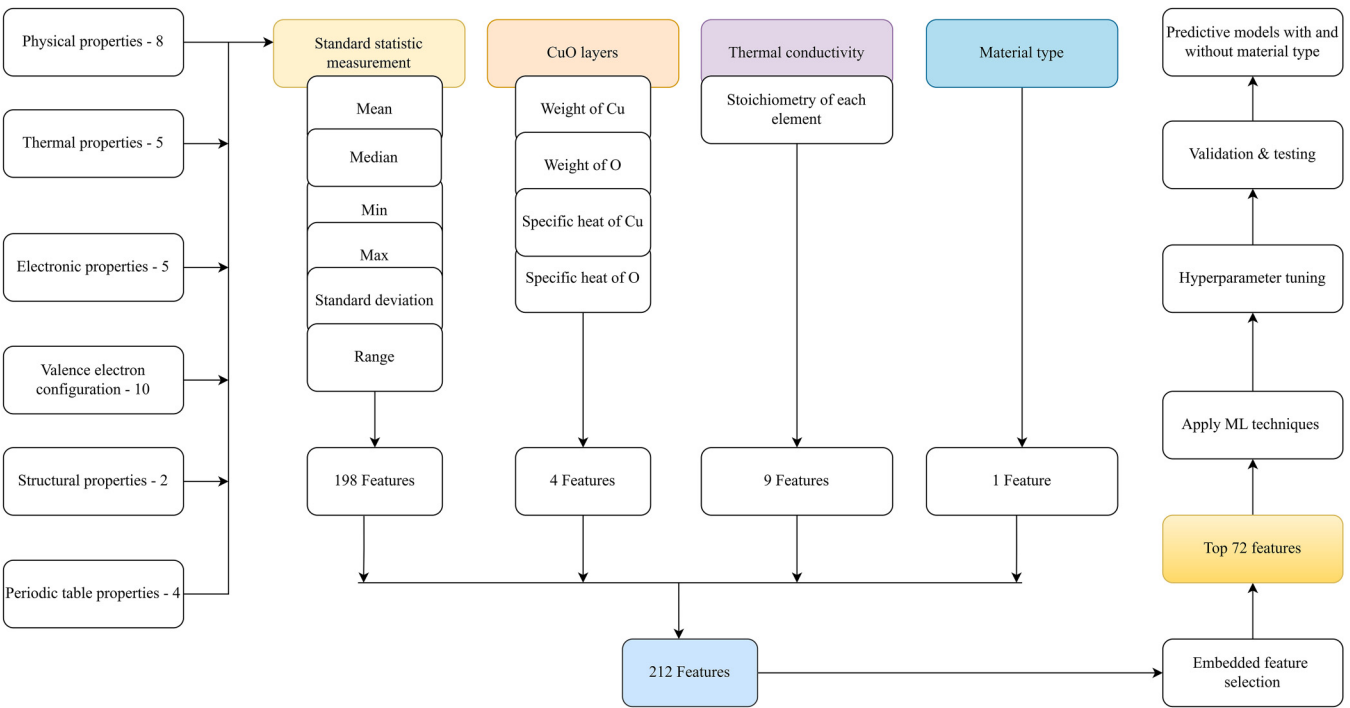
Machine-learning technique	Description
Multiple Regression	Used as a classical approach to model the relationship between a dependent variable and multiple independent variables.
Lasso Regression, Ridge Regression, and Elastic Net Regression	Utilized for their ability to handle multicollinearity and feature selection, enhancing model performance and interpretability.
Bagging	Reduces variance by averaging the results of multiple models trained on different subsets of the data.
Boosting	Reduces variance by averaging the results of multiple models trained on different subsets of the data.
Support vector machines (SVMs)	Known for their power in handling high-dimensional data making them suitable for our dataset with a large number of records.

features from an initial set of 212 features. Embedded methods combine the advantages of both filter-based and wrapper-based methods in terms of studying the importance of features while executing the ML model.³⁷ The RF model was selected due to its ability to model non-linear relationships and complex interactions among variables and robustness in handling high-dimensional datasets, owing to the ensemble-based structure that reduces bias and variance.

Feature importance was determined using the mean decrease in variance, calculated as the total reduction in node variance (weighted by the proportion of samples reaching each node) provided by the RF regression model. Features were ranked by their importance scores, and a threshold of cumulative feature importance exceeding 98% was applied to select the top 72 features. This threshold was chosen empirically to balance predictive power and dimensionality reduction, ensuring that the selected features contributed significantly to the model's performance. These 72 features were used in subsequent analyses to model the superconducting properties of the materials.

E. Machine learning approaches

To predict T_c of materials, we employed several widely used machine learning techniques, selected for their effectiveness in handling high-dimensional data and robustness to overfitting, as briefly summarized in Table II. These methods were implemented and



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FIG. 3. End-to-end flow chart detailing elemental feature processing, including categorized features (physical, thermal, electronic, valence, structural, and pertaining to the periodic table), statistical analyses yielding 198 features, additional CuO and material type features, the feature selection reducing to 72 key features, and development of final predictive models via machine learning with validation.

compared to identify the most effective approach for predicting T_c of materials, ensuring a balance between accuracy and model interpretability. Figure 3 summarizes the end-to-end flow of this work.

III. RESULTS AND DISCUSSION

Through the aforementioned feature selection procedure, 72 relevant features from the initial 212 were chosen. Normalization was performed on all data during the training and prediction processes. The entire dataset was randomly shuffled and split into training and test sets in an 8:2 ratio. The mean absolute error (MAE) in training, MAE in testing, and R^2 scores were recorded as evaluation criteria. The machine learning models described earlier (see Table II) were utilized in predicting T_c values while taking into account the potential for overfitting, which was not adequately considered in previous works that relied solely on R^2 values.

A. Identifying key features

During the feature selection process, an embedded method was employed to identify 72 of the most significant features. These features were subsequently ranked using the RF model, and the top 10 features are shown in Fig. 4.

It is evident that the type of material has a significant impact, as changes in material type often lead to variations in T_c .

Therefore, it is evident that relying solely on training models using data of only superconductors may not be sufficient or justifiable for accurately predicting T_c values for completely new materials. Since T_c can be influenced by factors beyond just the material's composition, such as structural arrangement and external conditions, it is crucial to consider a broader range of variables in the prediction task. Incorporating diverse datasets that encompass both superconductors and non-superconductors, along with relevant features such as material type, can enhance the model's ability to generalize and make accurate predictions for novel materials. This broader approach ensures that the model is better equipped to handle the complexities and nuances inherent in the superconductivity prediction task and is more suited for novel materials design.

Several prior investigations,^{14,40,41} alongside our study, underscore the significant influence of thermal conductivity in determining a material's T_c , which is ranked second in Fig. 4. Theoretical analyses propose that the thermal conductivity of superconductors offers profound insights into the behavior of their charge carriers, phonons, and the electron-phonon interactions.⁴² Thermal conductivity, representing a material's capacity to conduct heat, correlates directly with the density of particles capable of transferring thermal energy.^{42,43}

Out of the 212 features, a total of 72 features selected using the embedded method were utilized for both training and testing.

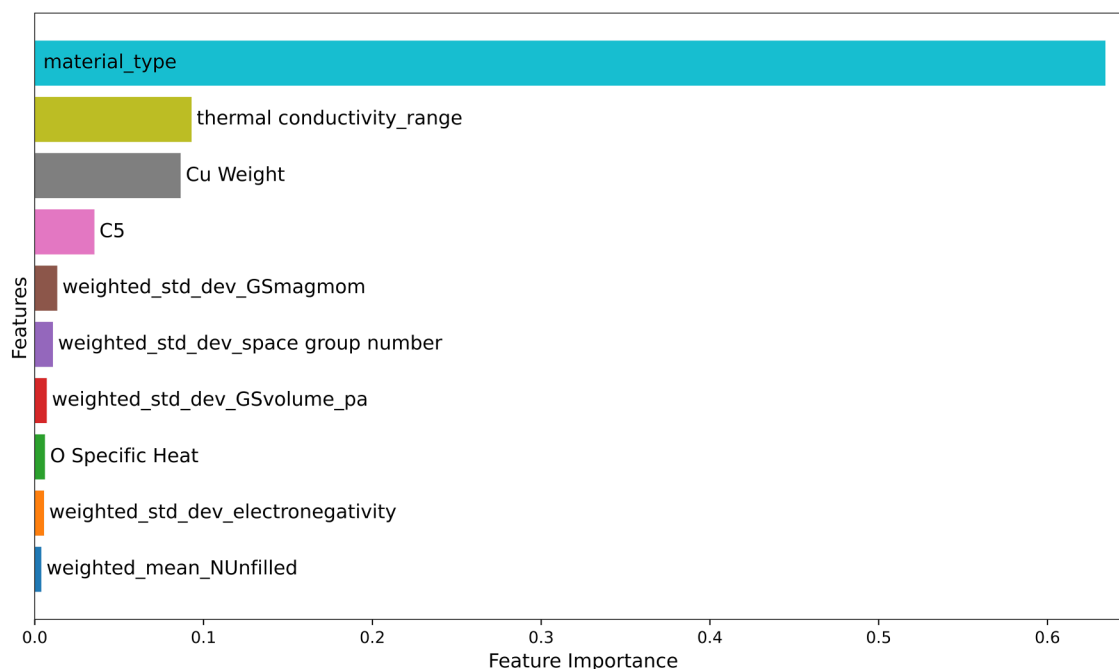


FIG. 4. The top 10 features identified using the embedded feature selection method. In the features shown, std_dev refers to the standard deviation, which measures the variability of data; GSvolume_pa denotes the ground-state volume per atom; GSmagmom denotes the ground-state magnetic moment; weighted_std_dev_GSmagmom represents the weighted standard deviation of the ground-state magnetic moment across all elements in a material calculated using the fractional abundance of each element in the composition; C5 denotes the thermal conductivity contribution of the fifth element in a material's composition; and NUnfilled denotes the number of unfilled electronic states.

TABLE III. The results of ML models on the training and testing datasets. For each ML model, MAE obtained in training and testing together with the R^2 value is summarized.

Model	Train MAE	Test MAE	R^2 value
Decision Tree	0.000	2.237	0.931
Extra Tree	0.000	2.219	0.931
Bagging Regression	0.739	1.862	0.958
AdaBoost Regression	4.233	4.264	0.883
XGBoost	1.296	2.230	0.957
LightGBM	1.849	2.280	0.955
Random Forest	0.656	1.748	0.962
Gradient Boosted	3.284	3.412	0.924
CatBoost	1.825	2.348	0.956
Linear Regression	7.615	7.645	0.789
Ridge Regression	7.613	7.643	0.789
Lasso Regression	8.042	8.049	0.754
Elastic Net Regression	8.112	8.121	0.716
SVR	4.554	4.646	0.798

Information pertaining to the complete set of 72 features can be made available upon request. The application of various machine learning techniques in determining T_c using this subset of features yielded diverse outcomes. The results of prediction accuracy and error are summarized in Table III.

In our analysis that addressed the issue of overfitting, the best-performing model achieved approximately 96% accuracy when using a RF model. However, upon closer examination, it became apparent that this model did not generalize well to unseen data due

to the significantly smaller MAE on the training set compared to the test set. When accounting for overfitting, the best models emerged as AdaBoost, Gradient Boosting, Logistic Regression (LR), Lasso Regression, and Elastic Net Regression. These models demonstrated better generalization capabilities and were thus deemed more suitable for our purpose. To enhance accuracy, hyperparameter tuning was employed through a grid search method. This involved systematically varying the parameters associated with each model and assessing their impact on accuracy. By iteratively adjusting these parameters and evaluating performance, the accuracy of each model was maximized, and the results are summarized in Table IV.

As can be observed from Table IV, parameter tuning in each model helps to eliminate the overfitting nature of the model (for instance, decision tree, extra tree, and random forest models in comparison to Table III). Among the tested models in this work, LGBM achieved an accuracy of 93%, indicating its effectiveness in handling the predictive task using our balanced and comprehensive dataset. As the literature reveals, LGBM is particularly well-suited to handle large datasets efficiently and is known for its capability to handle high-dimensional data and robustness against overfitting.⁴³ Additionally, LGBM's ability to optimize the decision boundaries and its flexibility in handling various types of features contribute to its superior performance in this context as well as many others. Overall, the combination of these factors makes LGBM an ideal choice for achieving high accuracy in predicting T_c of materials.³⁶

Table V lists a set of materials with known T_c that were not used for training the model, but were used only for testing the model's capability in predicting T_c of unseen materials. Table V includes both the actual and predicted T_c values generated using the LGBM model after hyperparameter tuning. It is clear that the

TABLE IV. The outcomes of training and testing of various ML models as a result of parameter tuning using a grid search. The parameter values that have been tuned in each ML model are also summarized.

Model	Parameters	Train MAE	Test MAE	R^2 value
Decision Tree	Depth—6	3.215	3.340	0.908
Extra Tree	Depth—7	3.117	3.194	0.920
	n_estimators—150			
Bagging Regression	Base estimator—Gradient Boosting	3.268	3.388	0.923
XGBoost	Learning_rate—0.04	3.055	3.333	0.912
	Depth—4			
LightGBM	boosting_types—Gradient Boosting	2.750	2.933	0.930
	Depth—4			
Random Forest	Depth—6	3.196	3.386	0.919
	n_estimator—200			
Gradient Boosted	Learning_rate—0.04	3.371	3.474	0.921
	n_estimator—200			
CatBoost	Learning_rate—0.03	3.034	3.280	0.915
AdaBoost	default	4.233	4.264	0.883
LR	default	7.615	7.645	0.789
Ridge	alpha = 0.01	7.614	7.645	0.789
Lasso	alpha = 0.01	7.604	7.624	0.789
Elastic net	alpha = 0.001	7.603	7.628	0.789
SVR	Kernel_types—rbf	3.690	3.784	0.853
	C_value—10			

TABLE V. Comparison of the actual and predicted T_c values using the best-performing LGBM model.

Material	Actual T_c (K)	Predicted T_c when material type is given (K)
Sm ₁ Se ₁	0.0	0.0
Ca ₁ Si ₁ Sr ₁	0.0	0.0
Ga _{1.02} Sb _{0.98} Te _{0.06}	0.0	0.0
Bi ₁ F ₆ Na ₁	0.0	0.0
Ge ₁ S ₁	0.0	0.0
Pt _{0.25} V _{0.75}	3.2	3.8
Mo ₁₃ Zr ₈₇	4.9	5.6
Ge ₁ Ta ₃	8.0	6.8
Th ₁ Co _{0.6} Ni _{0.4} C ₂	11.8	8.2
Bi ₂ Sr _{1.6} La _{0.4} Cu ₁ O _{6.35}	20.0	20.4
Ca ₁ Cs ₁ Fe ₄ As ₄	31.6	26.7
Nd _{1.4} Ba _{1.6} Cu ₃ O _{7.09}	38.0	39.9
Ca _{0.2} Ba _{1.55} La _{1.25} Cu ₃ O _{7.07}	46.1	50.1
La _{1.1} Ba _{1.4} Sr _{0.5} Cu ₃ O _{6.954}	50.1	50.5
Y _{0.8} Ca _{0.2} Ba _{1.4} La _{0.6} Cu ₃ O _{7.15}	66.0	56.8
Er _{0.7} Ca _{0.15} Hf _{0.15} Ba ₂ Cu ₃ O _{6.82}	77.3	83.9
Bi ₄ Ca ₃ Sr ₃ Cu ₄ O ₁₆	84.5	89.0
Y ₁ Ba ₂ Cu ₃ H _{0.08} O ₇	88.0	92.3
Tl ₁ Ba ₂ Ca ₂ Cu ₃ O ₉	110.0	107.8
Hg ₁ Ba ₂ Ca ₂ Cu ₃ O _{8.28}	118.5	121.9

predictions are very close to the actual values when prior knowledge on the material type, classified as HTS, LTS, and non-superconductor, was incorporated into the model.

It is evident that having prior knowledge of the material type or making an educated guess based on the class of materials one belongs to can significantly enhance the accuracy of predictions for new materials. Understanding the characteristics and properties associated with different material types enables tailoring the modeling approach more effectively. By incorporating information about the material type into the prediction task, such as non-superconductor, LTS, or HTS, the model can better account for the underlying factors influencing T_c . This prior knowledge about the material of focus serves as a valuable guide for the predictive models in achieving higher accuracy when predicting T_c of new materials, which may be branching out from classes or groups of materials for which properties are already known. Prior knowledge includes several key aspects, namely, elemental composition, crystal structure, properties such as density, melting point, and thermal conductivity of the material, and previous experimental results pertaining to T_c and trends of other physical properties observed in similar classes of materials.

Even without prior knowledge of the material type, the trained models can still provide insights into the range of temperatures at which a new material may exhibit superconductivity. By leveraging models trained on diverse materials using our dataset containing a variety of LTS, HTS, and non-superconductors, they can derive predictions, in terms of an interval for the T_c of new materials as

TABLE VI. Comparison of actual and predicted T_c values without prior knowledge about the material type. The best performing LGBM model is used to make predictions after hyperparameter tuning.

Material	Actual T_c (K)	Non-superconductor (class: 0)	LTS (class: 1)	HTS (class: 2)	Possible type	T_c range (K)
Sm ₁ Se ₁	0.0	0.0	3.28	89.65	0/1	(0–3.28)
Ca ₁ Si ₁ Sr ₁	0.0	0.0	7.13	88.09	0/1	(0–7.13)
Ga _{1.02} Sb _{0.98} Te _{0.06}	0.0	0.0	2.97	90.99	0/1	(0–2.97)
Bi ₁ F ₆ Na ₁	0.0	0.0	7.92	92.85	0/1	(0–7.92)
Ge ₁ S ₁	0.0	0.0	4.23	93.68	0/1	(0–4.23)
Pt _{0.25} V _{0.75}	3.2	0.0	3.77	94.27	0/1	(0–3.77)
Mo ₁₃ Zr ₈₇	4.9	1.05	5.61	86.07	0/1	(5.61–86.07)
Ge ₁ Ta ₃	8.0	1.79	6.84	95.56	0/1	(6.84–95.56)
Th ₁ Co _{0.6} Ni _{0.4} C ₂	11.8	2.85	8.22	100.59	0/1	(8.22–100.59)
Bi ₂ Sr _{1.6} La _{0.4} Cu ₁ O _{6.35}	20.0	0.60	20.48	79.23	0/1	(20.48–79.23)
Ca ₁ Cs ₁ Fe ₄ As ₄	31.6	8.63	26.69	91.12	0/1	(26.69–91.12)
Nd _{1.4} Ba _{1.6} Cu ₃ O _{7.09}	38.0	7.30	40.00	83.69	0/1	(40.00–83.69)
Ca _{0.2} Ba _{1.55} La _{1.25} Cu ₃ O _{7.07}	46.1	14.01	50.17	78.51	1/2	(50.17–78.51)
La _{1.1} Ba _{1.4} Sr _{0.5} Cu ₃ O _{6.954}	50.1	15.41	50.51	82.63	1/2	(50.51–82.63)
Y _{0.8} Ca _{0.2} Ba _{1.4} La _{0.6} Cu ₃ O _{7.15}	66.0	24.14	56.80	82.26	1/2	(56.80–82.26)
Er _{0.7} Ca _{0.15} Hf _{0.15} Ba ₂ Cu ₃ O _{6.82}	77.3	18.74	59.68	83.96	1/2	(59.68–83.96)
Bi ₄ Ca ₃ Sr ₃ Cu ₄ O ₁₆	84.5	21.67	61.47	89.02	1/2	(61.47–89.02)
Y ₁ Ba ₂ Cu ₃ H _{0.08} O ₇	88.0	19.94	60.03	92.39	1/2	(60.03–92.39)
Tl ₁ Ba ₂ Ca ₂ Cu ₃ O ₉	110.0	16.75	58.03	107.82	1/2	(58.03–107.82)
Hg ₁ Ba ₂ Ca ₂ Cu ₃ O _{8.28}	118.5	17.42	58.69	121.95	1/2	(58.69–121.95)

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shown in Table VI. These predictions provide valuable guidance by hinting at the potential material type and the corresponding range of temperature values where superconductivity may occur. Through this approach, one can effectively narrow down the search space for promising candidate materials and focus further experimental efforts or first-principles calculations on those most likely to exhibit promising superconducting behavior. This data-driven strategy enables informed decisions and streamlines the discovery process for novel superconducting materials, even in the absence of prior knowledge about the material type.

Below we discuss how to make interpretations of the outcome of the predictive model using two examples. The first example material is $\text{Hg}_1\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{8.25}$. Given the elements that make up this material, their properties, the structure of the CuO layers, an overview of the atomic structure, and data related to similar materials, one would recognize that $\text{Hg}_1\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{8.25}$ is likely an HTS, hence pick class 2 (suggested classes are 1 and 2) and its T_c would be the most plausible prediction. For this material, the predicted HTS (for class 2) T_c is 121.95 K, while the actual measured T_c is 118.5 K. However, even without detailed knowledge of the material's composition and properties, one can follow a general procedure to estimate its superconducting nature. The model generates three possible predicted values for the critical temperature, namely, non-

superconductor with predicted $T_c \approx 17.42$ K, LTS with predicted $T_c \approx 58.69$ K, or HTS with predicted $T_c \approx 121.95$ K. If the material is a non-superconductor, the predicted value should be close to zero. However, in this case, the predicted T_c for a non-superconductor (class 0) is 17.42 K, which deviates significantly from zero, indicating that the material is unlikely to be a non-superconductor. Given this information, it is safe to conclude that the material is likely a superconductor with T_c in the range 58.69–121.95 K.

The second example material is Sm_1Se_1 , where our model predicts three possible outcomes, namely, non-superconductor with predicted $T_c \approx 0$ K, LTS with predicted $T_c \approx 3.28$ K, or HTS with predicted $T_c \approx 89.65$ K. With prior knowledge about this material, it may be readily identified as a non-superconductor. The fact that the model yields a very low T_c (0 K) in the case of a non-superconductor further supports this view. In the absence of prior knowledge, comparing the three predicted values for the three possible types reveals that the first two values (non-superconductor and LTS classes) are much closer to each other, suggesting that the material's superconducting behavior, even if present, is likely within a very low-temperature range (0–3.28 K). For a comparison of actual and predicted T_c values across non-superconductors, LTS, and HTS, see Figs. 5 and 6.

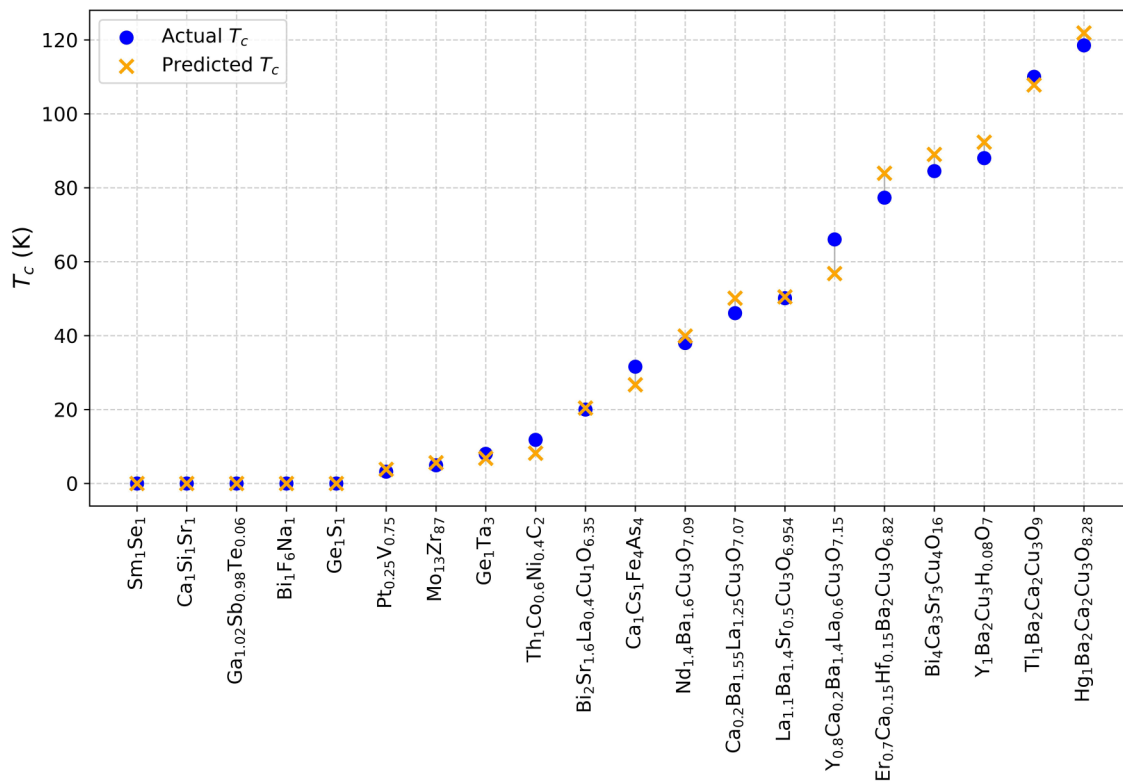


FIG. 5. Comparison of actual and predicted T_c values for several materials covering non-superconductors, LTS, and HTS.

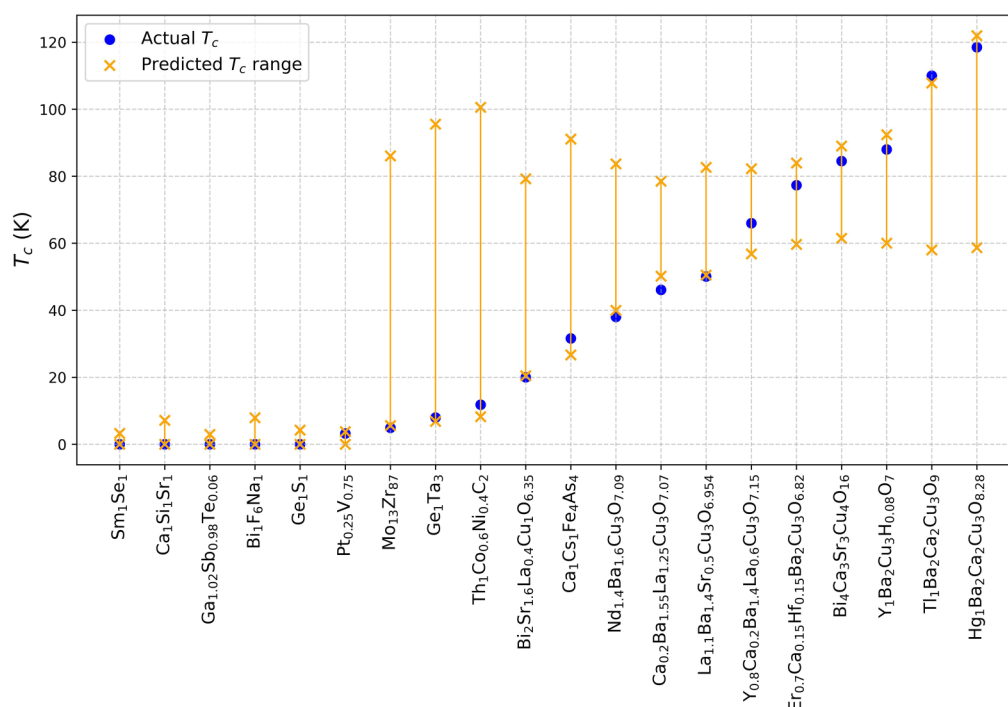


FIG. 6. Actual and predicted T_c range for several materials covering non-superconductors, LTS, and HTS.

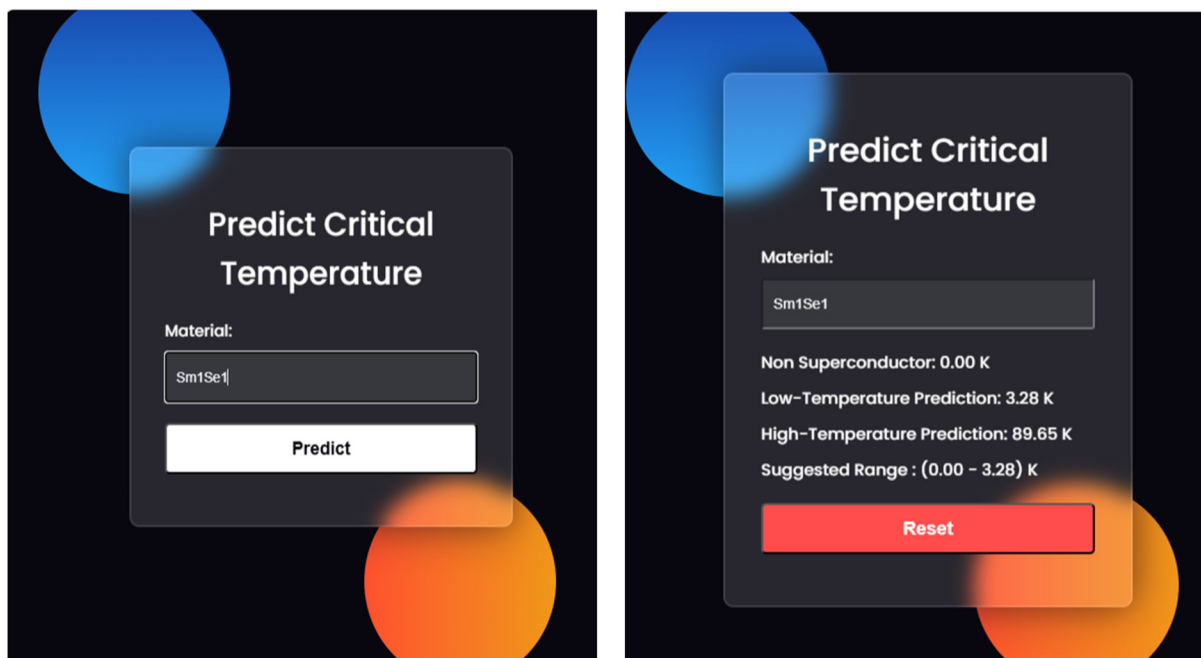


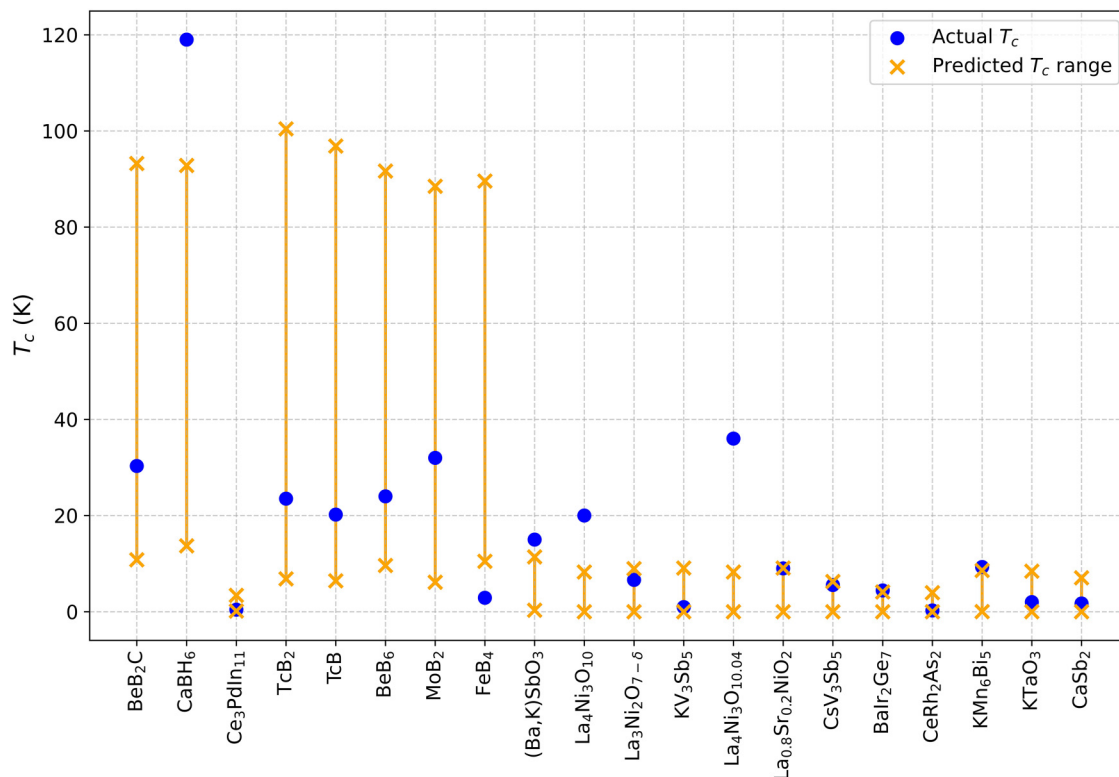
FIG. 7. Screenshot of the web application interface. When the material information is entered, the potential T_c as a non-superconductor, LTS, and HTS is presented together with the most plausible temperature range suggested by the algorithm. The user is able to make an educated pick following the procedure explained in the text.

TABLE VII. The table lists recently analyzed superconducting materials along with their T_c values and the corresponding predicted T_c ranges obtained using the selected final model.

Material	T_c (K)	Predicted range (K)
BeB ₂ C ⁴⁴	30.3	(10.80–93.24)
CaBH ₆ ⁴⁵	119	(13.70–92.81)
Ce ₃ PdIn ₁₁ ⁴⁶	0.42	(0.13–3.40)
TcB ₂ ⁴⁷	23.5	(6.83–100.44)
TcB ⁴⁸	20.2	(6.42–96.85)
BeB ₆ ⁴⁹	24 at ~4 GPa	(9.60–91.65)
MoB ₂ ⁵⁰	32 under 100 GPa	(6.13–88.49)
FeB ₄ ⁵¹	2.9	(10.49–89.58)
(Ba,K)SbO ₃ ⁵²	15	(0.30–11.41)
La ₄ Ni ₃ O ₁₀ ⁵³	20	(0.00–8.25)
La ₃ Ni ₂ O _{7-δ} ⁵⁴	6.6 at ~9 GPa Up to 41 at ~18 GPa	(0.00–8.96)
KV ₃ Sb ₅ ⁵⁵	0.93	(0.00–9.07)
La ₄ Ni ₃ O _{10.04} ⁵⁶	36 at ~48.0 GPa	(0.00–8.25)
La _{0.8} Sr _{0.2} NiO ₂ ⁵⁷	9	(0.00–9.08)
CsV ₃ Sb ₅ ⁵⁸	(3.5–7.6)	(0.00–6.28)
BaIr ₂ Ge ₇ ⁵⁹	4.4	(0.00–4.10)
CeRh ₂ As ₂ ⁶⁰	0.26	(0.00–3.95)
KMn ₆ Bi ₅ ⁶¹	9.3 at ~14.2 GPa	(0.01–8.56)
KTaO ₃ ⁶²	2	(0.00–8.42)
CaSb ₂ ⁶³	1.7	(0.00–7.04)

In further contributing to superconductivity research and materials design using materials informatics, we have developed a web application using the Flask (3.0.0) web framework written in Python that utilizes our trained model and the enhanced dataset for prediction tasks pertaining to potentially new materials as shown in Fig. 7. The application can be used to generate synthetic materials and predict their superconducting properties. By inputting the material compound (e.g., H₂O₁), users can estimate T_c to elucidate if it is a non-superconductor, LTS, or HTS. This tool allows users without access to a physical lab to predict T_c values based on ML models, providing insights into the superconducting potential of unknown materials. Based on the results, users may opt to perform first-principles calculations or actual experiments for promising candidates. In further solidifying the data-driven pipeline of the tool, it is useful to include basic features for performing molecular dynamics and/or crystal structure search based on first-principles calculations for materials of choice following the prediction task.

We finally perform a case study in trying to predict the T_c of some recent superconductors found in the literature,^{44–63} the data of which were not included in the comprehensive dataset we created initially for training and testing our models. The results are summarized in Table VII and Fig. 8, which show the expected T_c against the predicted one, highlighting the effectiveness of our dataset, models, and tool in searching for potential superconducting materials.

**FIG. 8.** Actual and predicted T_c range for new superconducting materials in recent literature, which are not included in the training or testing datasets for the models.

IV. CONCLUSIONS

In this work, we contributed to the data-driven initiative of superconductor research in two aspects. First, a new dataset comprising 13 415 superconducting compounds and 13 425 non-superconductors, encompassing 212 features was created to be used by researchers in this field. Second, several predictive models were trained and tested while paying close attention to overfitting to gauge their ability to predict the superconducting critical temperature of a given material when the type of material (non-superconductor, low-temperature superconductor, or high-temperature superconductor) is either known or unknown. The light gradient boosting model demonstrated the best performance under both situations. Our work has clearly exemplified that the inclusion of the additional features and diversification of the dataset result in increased in the predictive accuracy (93%). The novel idea of incorporating the material type enabled more accurate predictions of the superconducting critical temperature. Moreover, in cases where there was uncertainty about the material type, the model could suggest a range of superconducting critical temperatures for the material under consideration. Furthermore, the development of an easy to use web application presents a pioneering contribution to the field. This application facilitates the prediction and determination of the superconducting critical temperature of materials, thereby enhancing accessibility and usability of the dataset and models by the scientific community.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

P. A. H. Nawoda: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **Gayan P. Hettiarachchi:** Conceptualization (equal); Investigation (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **M. D. T. Attygalle:** Formal analysis (equal); Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Mendeley Data, at <https://scapp.pythonanywhere.com/>, Ref. 64.

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